MONTGOMERY'S ISOHOLONOMIC PROBLEM

The shortest loop with prescribed holonomy

An exact formula in every positive codimension

Take a k-plane in $\mathbb{C}^{k+r}$ and prescribe its unitary holonomy $W \in \mathrm{U}(k)$ under the canonical connection. Let $L_r(W)$ be the minimum length of a loop of planes realizing W, with projector speed $\|\dot{\Pi}\|_F/\sqrt{2}$. Sönnerborn's isoholonomic bound depends on the eigenphases of W. Section IV.B of the 2024 paper asks whether that bound remains tight with fewer auxiliary dimensions than nontrivial eigenphases [1].

Theorem. Write the eigenvalues of W as $e^{2\pi \mathrm{i}\alpha_j}$, with $1 > \alpha_1 \geq \cdots \geq \alpha_k \geq 0$, counted with multiplicity. For every $r \geq 1$,

$$\frac{L_r(W)^2}{(2\pi)^2} = \sum_{\substack{1 \leq j \leq i \leq k \\ i \equiv j \pmod r}} (1 - \alpha_j)\alpha_i. \tag{1}$$

A smooth minimizing loop is obtained by assigning the sorted phases to the r auxiliary directions in round-robin order. The $j = i$ terms are the published bound squared in these units. Equality in that bound holds exactly when $r \geq \operatorname{rank}(W - I)$.

The proof puts a minimizer in a form whose spectrum interlaces with the spectrum of its logical block. An energy comparison replaces each interlacing block by chains, one per auxiliary dimension. The remaining optimization balances the phase counts in every interval at once.

Prescribed holonomy	$r = 1$	$r = 2$	$r \geq 3$
$\mathrm{diag}(e^{3\pi \mathrm{i}/5}, e^{-3\pi \mathrm{i}/5})$	51/100	42/100	42/100
$\mathrm{diag}(-\mathrm{i}, -1, \mathrm{i}, 1)$	15/16	11/16	5/8
$-I_3$	3/2	1	3/4

Exact squared lengths in units of $(2\pi)^2$.

The theorem prescribes the full unitary, including its scalar phase. Section 5 also gives the exact answer for a projective target.

1 A global minimizer has a scalar auxiliary multiplier

Let V_0 be the initial orthonormal frame and let $V(t)$ be an H^1 frame with $V^*\dot{V} = 0$, $V(0) = V_0$, and $V(1) = V_0W$. For $\Pi = VV^*$, horizontality gives $\|\dot{\Pi}\|_F/\sqrt{2} = \|\dot{V}\|_F$. By constant-speed parametrization the infimum of the energy $E = \int_0^1 \|\dot{V}\|_F^2\, dt$ is $L_r(W)^2$. Section 4 constructs a competitor for every W. A bounded-energy sequence has a uniformly convergent subsequence with weakly convergent derivatives. The constraints pass to the limit, including $V^*\dot{V} = 0$ by uniform times weak convergence. Lower semicontinuity gives a global energy minimizer.

Identify V_0 with the first k coordinate vectors. Complete V by auxiliary columns B using $\dot{B} = -V\dot{V}^*B$ and the last r coordinate vectors at $t = 0$. Then $V^*B = 0$, $B^*B = I$, and $B^*\dot{B} = 0$. The completion $U = (V, B)$ has determinant 1 since its body velocity has trace zero. Its equation is $\dot{U} = UX$, where

$$X = \begin{pmatrix} 0 & C \\ -C^* & 0 \end{pmatrix}, \qquad \|X\|^2 = -\tfrac{1}{2}\operatorname{tr}(X^2) = \|C\|_F^2 = \|\dot{V}\|_F^2. \tag{2}$$

Let $\mathfrak{p}$ denote the off-diagonal space in (2) and $\mathfrak{k}$ the traceless skew-Hermitian block diagonal space. They are orthogonal for the trace pairing and $[\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}$.

For a minimizing control let S be the essential span of the ranges of $C(t)$. The evolution fixes $S^\perp$ in the logical plane since $X(t)$ annihilates it. Restriction to $S \oplus \mathbb{C}^r$ preserves minimality because competitors extend by the identity. The case $S = 0$ is constant. Work on the reduced space with full logical span.

The derivative of the endpoint map $X \mapsto V(1)$ in direction δX is $U(1)Y(1)V_0$, where $\dot{Y} = [Y, X] + \delta X$, $Y(0) = 0$. This follows by differentiating the integral equation for U, whose remainder is quadratic in $\|\delta X\|_{L^1}$. The map from $Y(1)$ to the frame tangent is onto because any frame tangent extends to a traceless skew-Hermitian body velocity. Its kernel is the traceless skew-Hermitian auxiliary block. A terminal covector for the frame endpoint is thus represented in the trace pairing by an $H(1)$ whose auxiliary block is scalar. Propagate it by $\dot{H} = [H, X]$. Invariance of the trace pairing gives

$$\langle H(1), Y(1)\rangle = \int_0^1 \langle H(t), \delta X(t)\rangle\, dt. \tag{3}$$

An annihilator of the control endpoint derivative would have $H(t) \in \mathfrak{k}$, hence $\dot{H} \in \mathfrak{p} \cap \mathfrak{k} = 0$. For constant $H = \operatorname{diag}(A, \mathrm{i}dI_r)$, commutation with $X(t)$ gives $AC(t) = \mathrm{i}dC(t)$. Full logical span and trace zero force $H = 0$. The endpoint derivative is therefore onto.

A multiplier for $\frac{1}{2}E$ gives $H = X + K$ by (3), with $K \in \mathfrak{k}$. The adjoint equation makes K constant and $\dot{X} = [K, X]$. Its auxiliary block is scalar by the terminal condition. For $P = X(0)$, the integrated trajectory is

$$U(t) = e^{t(P+K)}e^{-tK}. \tag{4}$$

2 Interlacing partitions the energy into chains

At the endpoint $U(1)$ is block diagonal. Thus $F = e^{P+K} = U(1)e^K$ is block diagonal. As F commutes with $P+K$, separation of the diagonal and off-diagonal blocks gives $[F,K] = [F,P] = 0$. Decompose into the eigenspaces of F. Each splits into a logical part and an auxiliary part. On an eigenspace with $F = e^{2\pi \mathrm{i}\gamma} I$, define

$$M = \frac{P+K}{2\pi \mathrm{i}} - \gamma I = \begin{pmatrix} A & Z \\ Z^* & bI_d \end{pmatrix}, \qquad Q = \frac{1}{2}\left(\sum_{\nu=1}^{s+d} n_\nu^2 - \sum_{\nu=1}^{s} a_\nu^2 - db^2 \right). \tag{5}$$

Here s, d are the two block sizes, $a_1 \le \cdots \le a_s$ are the eigenvalues of A, and $n_1 \le \cdots \le n_{s+d}$ are the integer eigenvalues of M. The auxiliary block is scalar by Section 1. The block contributes $Q = \|Z\|_F^2$ to $E/(2\pi)^2$ by tracing M^2 and using the constant speed of (4). Its logical endpoint eigenvalues are $e^{-2\pi \mathrm{i} a_\nu}$. A block with $d = 0$ contributes zero and fixes its logical part.

For $d > 0$, Cauchy interlacing gives $n_\nu \le a_\nu \le n_{\nu+d}$. Group indices by their residues modulo d. For each $h = 1, \ldots, d$ this gives an interlaced chain

$$n_h \;\le\; a_h \;\le\; n_{h+d} \;\le\; a_{h+d} \;\le\; n_{h+2d} \;\le\; \cdots .$$

Each chain ends at a full-spectrum point and can have zero logical points. Put $b_h = \sum_{\text{chain}} n - \sum_{\text{chain}} a$. The chains exhaust both spectra, so $\sum_h b_h = db$ by the trace of M. Hence the block energy bounds the sum of the chain costs

$$Q \;\ge\; \sum_{h=1}^{d} Q_h, \qquad Q_h = \frac{1}{2}\left(\sum_{\text{chain}} n^2 - \sum_{\text{chain}} a^2 - b_h^2 \right). \tag{6}$$

The difference is $\frac{1}{2}\sum_h (b_h - b)^2$.

Relabel a chain as $N_0 \le A_1 \le N_1 \le \cdots \le A_p \le N_p$. With $\ell_i = A_i - N_{i-1}$, $q_i = N_i - A_i$, we have $b_h = N_0 + \sum_i q_i$ and

$$Q_h = \sum_{1 \le j \le i \le p} \ell_j q_i. \tag{7}$$

To see this, use $N_i^2 - A_i^2 = 2N_i q_i - q_i^2$ and $N_i = N_0 + \sum_{j \le i}(\ell_j + q_j)$ in (6). For a phase fraction $\beta_i \in (0,1)$ with $A_i \equiv -\beta_i \bmod 1$, integer interlacing implies $\ell_i \ge 1 - \beta_i$ and $q_i \ge \beta_i$. Discard zero-phase positions from (7) by dropping their nonnegative terms. Replace the retained gaps by these lower bounds. The resulting sum is minimized in nonincreasing phase order. An adjacent swap of u, v changes only their mutual term, with old cost minus new cost $(1-u)v - (1-v)u = v - u$.

Summing (6) over the blocks partitions the nonzero phases of W into at most r groups. Each group contributes at least $\sum_{j \le i}(1-\beta_j)\beta_i$ for its own sorted phases.

3 The optimal partition balances every phase interval

For $x, y \in [0,1]$ the symmetric pair cost is

$$w(x,y) = \min(x,y)\big(1 - \max(x,y)\big) = \int_{0\le u\le v\le 1} \mathbf{1}_{\{x,y\in[u,v]\}}\, du\, dv. \tag{8}$$

The integral measures the intervals containing both points. The cost of a partition is the sum of w over unordered pairs in each group, including each phase paired with itself. For any interval $[u,v]$, let c_h count the phases in group h inside it. By (8) the cost of the partition is the integral of $\sum_h c_h(c_h+1)/2$.

For a fixed interval the total $\sum_h c_h$ is fixed. Transferring one count from a group with a to a group with $b \le a-2$ decreases the sum by $a-b-1$. The minimum therefore balances the counts to within one. An interval picks out a contiguous segment of the sorted phase list. The round-robin assignment $i \bmod r$ balances the counts on every contiguous segment, hence on every phase interval. It minimizes the integral and gives exactly the right-hand side of (1). This proves the lower bound for all horizontal loops.

4 Constructing a shortest loop

Assign one auxiliary vector to each nonempty round-robin group. For the group's sorted nonzero phases $\beta_1, \dots, \beta_p$, form the Hermitian matrix M and its diagonal D from

$$\begin{aligned} a_i &= i - \beta_i, \qquad b = \sum_{i=1}^{p} \beta_i, \qquad D = \operatorname{diag}(a_1, \dots, a_p, b), \\ M &= \begin{pmatrix} \operatorname{diag}(a_1,\dots,a_p) & z \\ z^* & b \end{pmatrix}, \qquad z_i = \sqrt{-\frac{\prod_{j=0}^{p}(a_i - j)}{\prod_{\ell\ne i}(a_i - a_\ell)}}. \end{aligned} \tag{9}$$

The strict interlacing $i-1 < a_i < i$ makes the radicands positive. At a_i the characteristic polynomial is $-|z_i|^2 \prod_{\ell\ne i}(a_i - a_\ell) = \prod_{j=0}^{p}(a_i - j)$. The polynomial $\prod_{j=0}^{p}(x-j)$ and the characteristic polynomial are monic with the same trace coefficient. Their difference has degree at most $p-1$ and p distinct zeros. Thus M has spectrum $0, \dots, p$.

The group's trajectory is

$$U_h(t) = e^{2\pi i t M} e^{-2\pi i t D}. \tag{10}$$

It is horizontal since its body velocity is $2\pi i e^{2\pi i t D}(M-D)e^{-2\pi i t D}$, and its determinant is 1 since $\operatorname{tr} M = \operatorname{tr} D$. At $t = 1$ the first exponential is the identity and the logical phases are exactly β_i. The projected loop is smooth and periodic. Its squared speed, in units of $(2\pi)^2$, is (7) with $\ell_i = 1 - \beta_i$, $q_i = \beta_i$. Run these trajectories in orthogonal subspaces and extend by the identity. The speed squares add and are constant, attaining (1).

5 Consequences and equality cases

The diagonal terms of (1) give the universal bound [1]. Put $m = \operatorname{rank}(W - I)$. If $r \geq m$, no off-diagonal term is positive. If $r < m$, the pair $j = 1,\ i = r+1$ contributes $(1-\alpha_1)\alpha_{r+1} > 0$. Thus the published bound is strict at the minimum when $r < m$. In particular, $L_1(\operatorname{diag}(e^{\mathrm{i}\theta}, e^{-\mathrm{i}\theta}))^2 = 4\pi\theta - \theta^2$ for $0 \leq \theta \leq \pi$.

When only the projective holonomy is fixed. Minimize (1) over scalar multiples of W, as in the projective formulation of the bound [7]. Between consecutive shifts at which an eigenvalue wraps to 1, all sorted phase fractions vary with slope one and their order is fixed. Each summand of (1) is a concave quadratic of the shift. At a wrap the terms involving the wrapping phases tend to zero from both sides, and the remaining index differences are preserved. Thus the expression is continuous and a minimum occurs at a wrap. It suffices to evaluate (1) on the at most k representatives that move one of the eigenvalues to 1.

All equality curves for the universal bound. Use a constant-speed horizontal lift in a holonomy eigenbasis. The j-th column becomes periodic after multiplying by $e^{-2\pi\mathrm{i}\alpha_j t}$, so it has twisted Fourier expansion $v_j(t) = \sum_{n\in\mathbb{Z}} c_{jn} e^{2\pi\mathrm{i}(n+\alpha_j)t}$. By Parseval, $p_{jn} = \|c_{jn}\|^2$ sum to one in each column. Integrated horizontality gives $\sum_n (n+\alpha_j)p_{jn} = 0$. Expanding $(n+\alpha_j)^2$ then gives the exact defect

$$\frac{E}{(2\pi)^2} = \sum_j \alpha_j(1-\alpha_j) + \sum_{j,n} n(n+1)p_{jn}. \tag{11}$$

Equality forces the modes $n = 0, -1$. Set $\Lambda = \operatorname{diag}(\alpha_1, \ldots, \alpha_k)$. Then $V(t) = (A + Be^{-2\pi\mathrm{i}t})e^{2\pi\mathrm{i}t\Lambda}$. The initial value and horizontality at $t = 0$ give

$$A = V_0(I - \Lambda) - Z, \qquad B = V_0\Lambda + Z, \qquad V_0^* Z = 0. \tag{12}$$

The coefficient of $e^{-2\pi\mathrm{i}t}$ in the orthonormality condition, after removing the diagonal phases, is $A^*B = 0$. It gives $Z^*Z = \Lambda(I - \Lambda)$. Conversely, any such Z gives $A^*A = I - \Lambda$, $B^*B = \Lambda$, and $A^*B = 0$. Substituting verifies the horizontal condition, constant speed and endpoint. The Gram identity describes the orthogonal auxiliary directions used by every saturating loop.

Literature and scope

Montgomery formulated the shortest-loop question for prescribed holonomy [2]. Sönnerborn's 2024 paper establishes the gate bound and the sufficient codimension threshold. Section IV.B explicitly asks about tightness below that threshold [1]. Earlier work constructs gate loops in ample codimension [3,4]. Tanimura, Hayashi and Nakahara also give codimension-one loop families and particular gate examples [5].

Albertini, D'Alessandro and Sheller use scalar exponentials and integer optimization for three-level control. Section IV of that work allows a scalar factor in assigning the target [6]. Here the scalar multiplier comes from the free auxiliary endpoint. It enables the trace comparison (6) and the exact formula for the full unitary in arbitrary codimension.

References

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The companion source archive constructs (9) with exact rational squared couplings when the phase fractions are rational. It checks the spectra by exact determinants, compares (1) with integer-spectrum interlacing data, and enumerates phase assignments to test the balancing argument.